

Cluster Naming Evaluation Report (Version 6)

OWNER vs. Span2Type: BERTScore Comparison

Five CrossNER Domains

Abstract

This report compares four cluster-naming methods across five CrossNER domains (AI, Literature, Music, Politics, Science) using BERTScore (RoBERTa-large) against confusion-matrix-derived ground-truth labels. All scores are measured at the **Entity Typing (ET) stage**. Two methods belong to **OWNER**: BERT-based MLM voting (avg $F_1=0.897$) and LLM-based naming via Ollama (avg $F_1=0.908$). Two methods belong to **Span2Type**, both using Maximum Marginal Relevance (MMR) exemplar selection: MMR-LLM naming (avg $F_1=0.924$, best overall) and MMR-BERT naming (avg $F_1=0.896$). All scores use the corrected ground truth established in Version 5. S2T-MMR-LLM is the strongest method; LLM-based methods consistently outperform MLM/BERT voting across all domains.

1 Introduction

Unsupervised NER systems such as OWNER [1] group entity spans into clusters labelled by integer IDs. Cluster naming assigns human-readable type labels without manual annotation. This report evaluates four naming strategies:

- **OWNER-BERT**: for each entity, fill ‘‘{sentence} {entity} is a [MASK].’’ with BERT-base-uncased; the most frequent top-1 token across the cluster is the name.
- **OWNER-LLM**: sample $n=16$ entities per cluster (uniform), query Llama-3.1-8B via Ollama (Figure 3 prompt, temperature = 0).
- **S2T-MMR-LLM**: uses *Maximum Marginal Relevance* (MMR) to select $k=5$ diverse, centroid-representative entities per cluster, then queries Llama-3.1-8B via Ollama with full example sentences (richer context); produces multi-word descriptive names.
- **S2T-MMR-BERT**: same MMR exemplar selection ($k=5$) as S2T-MMR-LLM, but uses BERT-base-uncased MLM filling instead of an LLM — the top-1 token from ‘‘{sentence} {entity} is a [MASK].’’ is the cluster name.

All methods are evaluated against ground-truth labels derived from the column-wise argmax of the entity-type confusion matrix ($\text{argmax}_i C_{ij}$), consistent with the corrected methodology of Version 5.

2 Experimental Setup

2.1 MLflow Run IDs

Table 1: MLflow run IDs for OWNER and Span2Type experiments.

Domain	OWNER	S2T-MMR-LLM	S2T-MMR-BERT	S2T-no-MMR
AI	ea6e40f8...eb6e	6b105207...2b2	9e28895d...ac9	22d48750...771
Literature	fb1215f7...f0	29e11a34...be	89ec5ac6...68b	a24b0559...dd
Music	2c886b9b...c5	56ad65e3...dc	d70574bc...b2	08d4cdde...2f
Politics	9c02f083...6d	610c691d...7b	5b3e1bf3...12	56aacc85...40
Science	f72db65d...96	efcfb115...55	eec16336...fe	14b05751...2b

2.2 Ground-Truth Alignment

For OWNER runs, GT labels are read from `et_test_cluster_naming_gt.json` (confusion-matrix argmax, corrected in Version 5). For Span2Type runs, GT labels are derived at runtime from `et_test_confusion-matrix.pt` using the same argmax procedure.

2.3 Cluster Statistics

Table 2: Number of clusters per domain and pipeline stage.

Domain	True k	OWNER / S2T-v1	S2T-MMR-BERT (NER)
AI	14	16	14
Literature	12	14	12
Music	13	20	20
Politics	9	22	22
Science	17	20	20
Total	65	92	88

3 Results

3.1 BERTScore F_1 Comparison

Table 3 presents BERTScore F_1 for all four methods at the **Entity Typing (ET)** stage. The best score per domain is shown in **bold**.

Table 3: BERTScore F_1 at the ET stage – four cluster-naming methods across five CrossNER domains. Best per domain in **bold**.

Domain	OW-BERT	OW-LLM	S2T-MMR-LLM	S2T-MMR-BERT
AI	0.900	0.892	0.922	0.900
Literature	0.891	0.923	0.913	0.889
Music	0.906	0.928	0.943	0.911
Politics	0.892	0.910	0.928	0.887
Science	0.894	0.886	0.912	0.893
Avg	0.897	0.908	0.924	0.896

Overall ranking (ET stage):

S2T-MMR-LLM (**0.924**) > OW-LLM (0.908) > OW-BERT $\approx$ S2T-MMR-BERT (0.897 / 0.896)

3.2 S2T-MMR-BERT Detailed Scores (P / R / F_1)

Table 4 shows full BERTScore Precision, Recall, and F_1 for S2T-MMR-BERT at the Entity Typing (ET) stage.

3.3 S2T-MMR-LLM Detailed Scores (P / R / F_1)

Table 5 shows full BERTScore Precision, Recall, and F_1 for S2T-MMR-LLM at the Entity Typing (ET) stage.

3.4 Qualitative Naming Examples

Table 6 shows representative S2T-MMR-BERT predictions with per-cluster BERTScore F_1 .

Table 4: S2T-MMR-BERT BERTScore at the ET stage (P / R / F₁).

Domain	ET Stage		
	P	R	F ₁
AI	0.897	0.904	0.900
Literature	0.883	0.896	0.889
Music	0.906	0.916	0.911
Politics	0.880	0.894	0.887
Science	0.893	0.893	0.893
Avg	0.892	0.900	0.896

Table 5: S2T-MMR-LLM BERTScore at the ET stage (P / R / F₁).

Domain	ET Stage		
	P	R	F ₁
AI	0.919	0.925	0.922
Literature	0.909	0.917	0.913
Music	0.940	0.946	0.943
Politics	0.925	0.931	0.928
Science	0.913	0.911	0.912
Avg	0.921	0.926	0.924

4 MMR Ablation

To isolate the effect of Maximum Marginal Relevance exemplar selection, we compare two Span2Type configurations: **no-MMR** ($k=16$) versus **MMR** ($k=5$ diverse exemplars). The no-MMR run IDs are listed in the rightmost column of Table 1.

4.1 Ollama Backend

With the Ollama backend, MMR improves average F₁ by +0.003 overall, but the gain is domain-dependent. Literature and Science benefit most (+0.013, +0.007), where the no-MMR uniform sample yields redundant entities that lead the LLM to produce vague labels (*Book*, *Science*); MMR selects diverse centroid-representative exemplars, eliciting more specific outputs (*Literary work*, *Scientific discipline*). Conversely, Music and Politics show a small regression (−0.003, −0.001): no-MMR already samples a broad mix of band names, song titles, and politician names that give the LLM sufficient signal, while MMR occasionally over-concentrates on centroid entities, slightly narrowing the context. The AI domain is effectively tied (+0.0001), consistent with abstract technical types being hard for both sampling strategies.

4.2 BERT Backend

With the BERT backend, MMR yields a small but consistent improvement on average (+0.003), with the largest gain in Music (+0.016): the no-MMR uniform sample includes many redundant entity mentions from a few dominant clusters (e.g., several occurrences of the same band name), causing the MLM head to converge on generic tokens; MMR diversifies the exemplar set, shifting the argmax towards more cluster-specific tokens (*band*, *musician*, *genre*). AI and Science show minor regressions (−0.004, −0.001) because MMR’s centroid-representative selection occasionally surfaces technically-worded entity spans where the MLM’s most frequent token is less aligned with the GT label than the token obtained under uniform sampling. Crucially, MMR cannot suppress the **vegetarian** artefact: the noise prediction persists in 3–5 clusters per domain regardless of

Table 6: Selected S2T-MMR-BERT cluster naming examples. **Green** = exact/near-exact match; **orange** = semantically close; **red** = incorrect or noise.

Dom.	GT	Prediction	Quality	F ₁
AI	university	university	Exact	1.000
AI	conference	conference	Exact	1.000
AI	task	problem	Close	0.943
AI	algorithm	standard	Partial	0.908
AI	researcher	physicist	Close	0.858
AI	programlang	standard	Off	0.838
AI	person	vegetarian	Noise	0.853
Lit.	book	novel	Close	0.906
Lit.	event	festival	Close	0.918
Lit.	country	nation	Close	0.924
Lit.	writer	poet	Close	0.897
Lit.	award	retrospective	Partial	0.865
Lit.	book	vegetarian	Noise	0.843
Mus.	country	country	Exact	1.000
Mus.	band	band	Exact	1.000
Mus.	musicgenre	genre	Close	0.936
Mus.	musicalartist	musician	Close	0.930
Mus.	award	nomination	Close	0.897
Mus.	band	vegetarian	Noise	0.848
Pol.	politicalparty	party	Close	0.930
Pol.	location	city	Close	0.937
Pol.	politician	democrat	Close	0.899
Pol.	person	vegetarian	Noise	0.853
Sci.	university	university	Exact	1.000
Sci.	protein	protein	Exact	1.000
Sci.	scientist	physicist	Close	0.935
Sci.	chemicalcompound	catalyst	Partial	0.881
Sci.	chemicalelement	semiconductor	Partial	0.834
Sci.	person	vegetarian	Noise	0.853

exemplar selection strategy, confirming that the limitation is intrinsic to greedy MLM top-1 decoding.

4.3 Ablation Summary

Both backends improve by $\approx +0.003$ from lowest to highest configuration. For the Ollama backend the gain requires MMR selection, MMR elicits richer, more plural-inflected outputs from the LLM; . For the BERT backend, the $+0.003$ gain comes entirely from MMR selection concentrating the MLM prompt on diverse, centroid-representative exemplars.

5 Analysis

5.1 S2T-MMR-LLM Is the Strongest Method

S2T-MMR-LLM (avg F₁=0.924) outperforms all other methods in four of five domains. It combines MMR exemplar selection with rich sentence context passed to Llama-3.1-8B, enabling the LLM to infer fine-grained types such as *machine learning library* (GT: product), *legislative body* (GT: organisation), and *biological processes* (GT: misc).

Table 7: S2T Ollama: no-MMR ($k=16$, no lem) vs MMR ($k=5$, +lem). Best per domain in **bold**.

Domain	no-MMR	MMR	Δ
AI	0.9218	0.9219	+0.0001
Literature	0.9063	0.9197	+0.0134
Music	0.9539	0.9509	−0.0030
Politics	0.9302	0.9293	−0.0009
Science	0.9084	0.9156	+0.0072
Avg	0.9241	0.9275	+0.0034

Table 8: S2T BERT: no-MMR ($k=16$) vs MMR ($k=5$). ; Δ reflects MMR. Best per domain in **bold**.

Domain	no-MMR	MMR	Δ
AI	0.9037	0.9000	−0.0037
Literature	0.8874	0.8888	+0.0014
Music	0.8948	0.9110	+0.0162
Politics	0.8830	0.8868	+0.0038
Science	0.8935	0.8925	−0.0010
Avg	0.8925	0.8958	+0.0033

5.2 LLM Back-end Drives the Quality Gap, Not MMR Alone

Both S2T methods use the same MMR exemplar selection ($k=5$), yet S2T-MMR-LLM (0.924) substantially outperforms S2T-MMR-BERT (0.896). This +0.028 gap is attributable entirely to the naming back-end: LLM outputs are multi-word, context-aware descriptions, whereas BERT top-1 token voting collapses to a single common token. Comparing S2T-MMR-BERT with OWNER-BERT (0.897) — the latter without MMR — shows that MMR selection itself gives no measurable benefit (−0.001), confirming that the bottleneck for MLM naming lies in the vocabulary and greedy decoding of the MLM head, not in exemplar selection.

5.3 LLM Methods Outperform MLM Voting

Both LLM methods (OW-LLM: 0.908, S2T-MMR-LLM: 0.924) score above both MLM methods (OW-BERT: 0.897, S2T-MMR-BERT: 0.896). MLM top-1 voting is limited to single-token outputs and frequently falls back to degenerate predictions (**vegetarian**) for ambiguous clusters. LLM methods produce multi-word names with stronger semantic alignment to CrossNER entity type labels.

5.4 The vegetarian Artefact

The **vegetarian** noise token appears consistently across all MLM methods: 4–5 clusters per domain (out of 14–22) receive this prediction. It is the most impactful single failure mode for BERT/MLM-based naming, producing per-cluster $F_1 \approx 0.85$ against GT labels such as **person**, **book**, or **band**. LLM methods are immune to this artefact.

5.5 Per-Domain Observations

- **AI**: All methods score 0.89–0.92. Abstract types (**algorithm**, **programminglang**, **task**) are hard for all methods; only S2T-MMR-LLM captures *programming languages* and *machine learning* as reasonable proxies.

Table 9: MMR ablation summary: lowest (no-MMR) vs highest (MMR) per backend.

Backend	no-MMR	MMR	Δ
Ollama	0.9241	0.9275	+0.0034
BERT	0.8925	0.8958	+0.0033

- **Literature:** OW-LLM wins (0.923) due to descriptive names (*Novel, Author, Film Festival*) that align well with CrossNER types.
- **Music:** S2T-MMR-LLM dominates (0.943). Three exact matches for S2T-MMR-BERT (*band, band, country*) keep it competitive (0.911).
- **Politics:** S2T-MMR-LLM best (0.928) via *political party, legislative body, politician*. BERT methods predict generic tokens (*party, democrat*).
- **Science:** S2T-MMR-LLM leads (0.912); S2T-MMR-BERT recovers with exact matches (*university, protein*). Science-specific compound types (*chemicalelement, academicjournal*) remain hard for all methods.

6 Conclusion

The four-method ranking at the ET stage is:

S2T-MMR-LLM (**0.924**) > OW-LLM (0.908) > OW-BERT (0.897) $\approx$ S2T-MMR-BERT (0.896)

The MMR ablation (Section 4) shows that both backends gain $\approx +0.003$ when moving from the no-MMR baseline to the best MMR. Across all conditions, the primary quality driver remains the naming back-end: LLM methods outperform MLM methods by 0.028–0.032 regardless of exemplar selection strategy. The *vegetarian* artefact remains the main failure mode for all MLM/BERT approaches and cannot be resolved without filtering heuristics or switching to an LLM-based back-end.

References

- [1] P.-Y. Genest, P.-E. Portier, E. Egyed-Zsigmond, and M. Lovisetto, “OWNER – Toward Unsupervised Open-World Named Entity Recognition,” *IEEE Access*, vol. 13, pp. 50077–50096, 2025.
- [2] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, and Y. Artzi, “BERTScore: Evaluating Text Generation with BERT,” in *Proc. ICLR*, 2020.